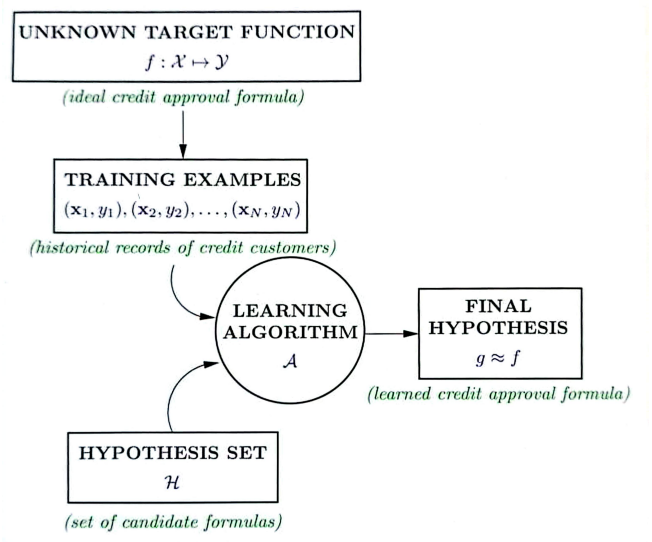
The Australian National University
School of Computing
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Semester 2, 2026
Tutorial Week 2

COMP3670 / COMP6670: Introduction to Machine Learning

Question 1 The supervised learning framework

Recall the supervised learning framework from the lecture:



Source: Abu-Mostafa, Y., Magdon-Ismail, M., & Lin, H.-T. (2012). *Learning from Data.* AMLBook. Figure 1.2.

Express each of the following tasks in the framework of learning from data by specifying the input space $\mathcal{X}$, output space $\mathcal{Y}$, target function $f : \mathcal{X} \to \mathcal{Y}$, and the specifics of the dataset that we will learn from.

(a) Medical diagnosis: A patient walks in with a medical history and some symptoms, and you want to identify the problem.

(b) Handwritten digit recognition (for example postal zip code recognition for mail sorting).

(c) Determining if an email is spam or not.

(d) Predicting how an electric load varies with price, temperature, and day of the week.

(e) A problem of interest to you for which there is no analytic solution, but you have data from which to construct an empirical solution.

Question 2 The Perceptron for Spam Detection

Suppose that we use a perceptron to detect spam messages. Let's say that each email message is represented by the frequency of occurrence of keywords, and the output is +1 if the message is considered spam.

(a) Write down in mathematical notation the feature vector $\mathbf{x}$ and its domain $\mathcal{X}$. Then write down the formula for the perceptron output $h_{\mathbf{w}}(\mathbf{x})$.

(b) Can you think of some keywords that will end up with a large positive weight in the perceptron?

(c) How about keywords that will get a negative weight?

(d) What parameter in the perceptron directly affects how many borderline messages end up being classified as spam?

Question 3 **The Perceptron Learning Algorithm**

The **Perceptron Learning Algorithm (PLA)** assumes the dataset is linearly separable and proceeds as follows:

1. **Initialise**: Set $w_0(0) = w_1(0) = \cdots = w_n(0) = 0$.
2. For each iteration $t = 1, 2, \ldots$:
   - Find any misclassified training example $(\mathbf{x}_i, y_i)$ and denote it by $(\mathbf{x}(t), y(t))$ the example selected at step t.
   - **Update** the weights, for $j = 1, \ldots, n$:
   $$w_j(t) \leftarrow w_j(t-1) + y(t)\, x_j(t), \qquad w_0(t) \leftarrow w_0(t-1) + y(t). \tag{1}$$
3. **Repeat** step 2 until no misclassified example remains.

The perceptron classifies $\mathbf{x}(t)$ according to $\text{sign}\Big(\sum_{j=1}^{n} w_j(t-1)\, x_j(t) + w_0(t-1)\Big)$, and the weight update rule in (1) has the nice interpretation that it moves in the direction of classifying $\mathbf{x}(t)$ correctly. In this exercise you will show this nice property.

(a) Show that $y(t)\left(\sum_{j=1}^{n} w_j(t-1)\, x_j(t) + w_0(t-1)\right) < 0$. [*Hint*: $\mathbf{x}(t)$ is misclassified by $\mathbf{w}(t-1)$.]

(b) Show that

$$y(t)\left(\sum_{j=1}^{n} w_j(t)\, x_j(t) + w_0(t)\right) > y(t)\left(\sum_{j=1}^{n} w_j(t-1)\, x_j(t) + w_0(t-1)\right).$$

[*Hint*: Use (1).]

(c) As far as classifying $\mathbf{x}(t)$ is concerned, argue that the move from $\mathbf{w}(t-1)$ to $\mathbf{w}(t)$ is a move 'in the right direction'.

For simplification, assume throughout this exercise that $\sum_{j=1}^{n} w_j(t-1)\, x_j(t) + w_0(t-1) \neq 0$ for all t.